3.3 Problems

$u_{tt} = c^2 u_{xx}$ $\quad u(0,t) = u(L,t) = 0$ $\quad$ c + L to be specified in the problems.

$u(x,0) = f(x)$ $\quad u_t = (x,0) = g(x)$

a.) $f(x) = \sin(x)\cos(x)$, $g(x) = 0$, $c = 1$, $L = \pi$

$$u_{tt} = c^2 u_{xx} \Rightarrow u_{tt} = u_{xx}$$

$$\frac{T''(t)}{T(t)} = \lambda = \frac{X''(x)}{X(x)} \left\} \begin{array}{l} X''(x) - \lambda X = 0 \\ T''(t) - \lambda T(t) = 0 \end{array} \right.$$

1) $\lambda > 0$ $\quad X(x) = D_1 e^{-\sqrt{\lambda}x} + D_2 e^{\sqrt{\lambda}x}$

$X(0) = D_1 + D_2 = 0 \rightarrow D_1 = -D_2 / -D_1 = D_2$

$X(x) = D_1 e^{-\sqrt{\lambda}x} - D_1 e^{\sqrt{\lambda}x}$

$0 = D_1 - D_1 \rightarrow$ Trivial

2) $\lambda = 0$ $\quad X(x) = D_1 + D_2 x$

$$\left.\begin{array}{l} X(0) = D_1 = 0 \,/\, D_1 = 0 \\ X(L) = D_2 L = 0 \,/\, D_2 = 0 \end{array}\right\} \text{Trivial}$$

3.) $\lambda < 0$ $\quad X_n(x) = D_{2,n} \sin\left(\frac{n\pi x}{L}\right) / T(t) = b_{1,n}\cos(c_n t) + b_{2,n}\sin(c_n t)$

$u_n(x,t) = \sin\left(\frac{n\pi x}{L}\right)\left[b_{1,n}\cos(c_n t) + b_{2,n}\sin(c_n t)\right]$ $(L = \pi)$

$u_n(x,t) = \sin(nx)\left[b_{1,n}\cos(c_n t) + b_{2,n}\sin(c_n t)\right]$

$u(x,t) = \sum_{n=1}^{\infty} \sin(nx)\left[b_{1,n}\cos(c_n t) + b_{2,n}\sin(c_n t)\right]$

$$u(x,0)=\sum_{n=1}^{\infty}\sin(nx)\left[b_{1,n}\right] \quad \text{on } 0\leq x\leq \pi$$

Recall: $f(x)=\sin(x)\cdot\cos(x)$ / $\sin(2x)=2\sin(x)\cos(x)$

$=\frac{1}{2}\sin(2x)$

$$b_{1,n}=\frac{2}{\pi}\int_0^{\pi} f(x)\cdot\sin(nx)\,dx=\frac{2}{\pi}\int_0^{\pi}\frac{1}{2}\sin(2x)\cdot\sin(nx)\,dx$$

$=\frac{1}{2}$ if $n=2$, 0 otherwise

$$\frac{\partial}{\partial t}u(x,t)=\sum_{n=1}^{\infty}\sin(nx)\left[-b_{1,n}\cdot c_n\sin(c_n t)+b_{2,n}\cdot c_n\cdot\cos(c_n t)\right]$$

Recall: $u_t(x,0)=g(x)=0$ Thus...

$$u_t(x,0)=\sum_{n=1}^{\infty}\sin(nx)\left[b_{2,n}\cdot c_n\right]=0 \quad \therefore b_{2,n}=0$$

So, $u(x,t)=\sum_{n=1}^{\infty}\sin(nx)\left[b_{1,n}\cos(c_n t)\right]$ where $b_{1,n}$ was defined above. Q.E.D.

b.) $f(x) = 2\sin(\pi x)$, $g(x) = 0$, $c = \frac{1}{\pi}$, $L = 1$

$u_{tt} = c^2 u_{xx} \Rightarrow u_{tt} = \frac{1}{\pi^2} u_{xx}$

$$\frac{T''(t)\cdot\pi^2}{T(t)} = \lambda = \frac{X''(x)}{X(x)} \left\{ \begin{array}{l} T''(t)\cdot\pi^2 - \lambda T(t) = 0 \\ X''(x) - X(x)\cdot\lambda = 0 \end{array} \right.$$

1) $\lambda > 0$ $\quad X(x) = D_1 e^{-\sqrt{\lambda}x} + D_2 e^{\sqrt{\lambda}x}$

$X(0) = D_1 + D_2 = 0 \rightarrow D_1 = -D_2 / -D_1 = D_2$

$X(x) = D_1 e^{-\sqrt{\lambda}x} - D_1 e^{\sqrt{\lambda}x}$

$X(0) = D_1 - D_1 = 0 \rightarrow$ Trivial

2.) $\lambda = 0$ $\quad X(x) = D_1 + D_2 x$

$$\left. \begin{array}{l} X(0) = D_1 = 0 \rightarrow D_1 = 0 \\ X(L) = D_2 L = 0 \rightarrow D_2 = 0 \end{array} \right\} \text{Trivial}$$

3) $\lambda < 0$ $\quad X_n(x) = D_{2,n} \sin\left(\frac{n\pi x}{L}\right) / T(t) = E_n \cos(c_n t) + F_n \sin(c_n t)$

$u_n(x,t) \xrightarrow{L=1} \sin(n\pi x)\left[E_n \cos(c_n t) + F_n \sin(c_n t)\right]$

$$u(x,t) = \sum_{n=1}^{\infty} \sin(n\pi x)\left[E_n \cos(c_n t) + F_n \sin(c_n t)\right]$$

$$u(x,0) = \sum_{n=1}^{\infty} \sin(n\pi x)\left[E_n\right]$$ (E_n is equivalent to $b_{1,n}$)

$$b_{1,n} = E_n = \frac{2}{\pi}\int_0^1 f(x)\cdot\sin(n\pi x)\,dx = \frac{4}{\pi}\int_0^1 \sin(\pi x)\sin(n\pi x)\,dx$$

Cont...

$b_{1,n} = E_n = \frac{2}{\pi}$ if $n=1$; 0 otherwise

$$u_t(x,t) = \sum_{n=1}^{\infty} \sin(n\pi x)\left[-E_n \cdot c_n \sin(c_n t) + F_n \cdot c_n \cdot \cos(c_n t)\right]$$

Recall: $u_t(x,0) = g(x) = 0$ Thus...

$$u_t(x,0) = \left\{\sum_{n=1}^{\infty} \sin(n\pi x)\left[F_n \cdot c_n\right]\right\} = 0 \quad \therefore F_n = b_{2,n} = 0$$

so, $u(x,t) = \sum_{n=1}^{\infty} \sin(n\pi x)\left[E_n \cos(c_n \cdot t)\right]$ where E_n was defined above. Q.E.D.

c.) $f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ (1-x) & 1 < x \leq 2 \end{cases}$ $g(x) = 0,\ c = 4,\ L = 2$

$u_{tt} = c^2 u_{xx} \Rightarrow u_{tt} = 16 \cdot u_{xx}$

$$\frac{T''(t)}{16\,T(t)} = \lambda = \frac{X''(x)}{X(x)} \Bigg\} \begin{array}{l} T''(t) - 16\lambda T(t) = 0 \\ X''(x) - X(x)\lambda = 0 \end{array}$$

1) $\lambda > 0$ $X(x) = D_1 e^{-\sqrt{\lambda}x} + D_2 e^{\sqrt{\lambda}x}$

$X(0) = D_1 + D_2 = 0 \rightarrow D_1 = -D_2 / -D_1 = D_2$

$X(x) = D_1 e^{-\sqrt{\lambda}x} - D_1 e^{\sqrt{\lambda}x}$

$X(0) = D_1 - D_1 = 0 \rightarrow$ Trivial

2) $\lambda = 0$ $X(x) = D_1 + D_2 x$

$$\left.\begin{array}{l} X(0) = D_1 = 0 \rightarrow D_1 = 0 \\ X(L) = D_2(L) = 0 \rightarrow D_2 = 0 \end{array}\right\} \text{Trivial}$$

3) $\lambda < 0$ $X_n(x) = D_{2,n} \sin\left(\frac{n\pi x}{L}\right) / T(t) = b_{1,n} \cos(c_n t) + b_{2,n} \sin(c_n t)$

$u_n(x,t) \xrightarrow{L=2} D_{2,n} \sin\left(\frac{n\pi x}{2}\right)\left[b_{1,n} \cos(c_n t) + b_{2,n} \sin(c_n t)\right]$

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{2}\right)\left[b_{1,n} \cos(c_n t) + b_{2,n} \sin(c_n t)\right]$$

$$u(x,0) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{2}\right)\left[b_{1,n}\right] \ ;\ b_{1,n} = \frac{2}{(2)} \int_0^2 f(x) \cdot \sin\left(\frac{n\pi x}{2}\right) dx \ *$$

* This is the general formula. Since $f(x)$ is a piecewise function, we must use 2 separate formulas.

Cont...

α: $f(x) = x$ on $0 \le x \le 1$

$$\int_0^1 x \cdot \sin\left(\frac{n\pi x}{2}\right) dx$$

$+x$	$\sin\left(\frac{n\pi x}{2}\right)$
-1	$-\frac{2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)$
$+0$	$-\frac{4}{n^2\pi^2}\sin\left(\frac{n\pi x}{2}\right)$

$$= -x \cdot \frac{2}{n\pi} \cdot \cos\left(\frac{n\pi x}{2}\right) + \frac{4}{n^2\pi^2}\sin\left(\frac{n\pi x}{2}\right)\Big|_0^1$$

$$\left\{\left[-\frac{2}{n\pi} \cdot \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} \cdot \sin\left(\frac{n\pi}{2}\right)\right] - [0]\right\}$$

$$= \frac{-2n\pi \cdot \cos\left(\frac{n\pi}{2}\right) + 4\sin\left(\frac{n\pi}{2}\right)}{n^2\pi^2}$$

β: $f(x) = 1 - x$ on $1 \le x \le 2$

$$\int_1^2 (1-x) \cdot \sin\left(\frac{n\pi x}{2}\right) dx$$

$+(1-x)$	$\sin\left(\frac{n\pi x}{2}\right)$
$-\ -1$	$-\frac{2}{n\pi} \cdot \cos\left(\frac{n\pi x}{2}\right)$
$+\ 0$	$-\frac{4}{n^2\pi^2} \cdot \sin\left(\frac{n\pi x}{2}\right)$

$$= -(1-x) \cdot \frac{2}{n\pi} \cdot \cos\left(\frac{n\pi x}{2}\right) - \frac{4}{n^2\pi^2}\sin\left(\frac{n\pi x}{2}\right)$$

$$\left\{\left[\frac{2}{n\pi} \cdot \cos(n\pi) - \frac{4}{n^2\pi^2} \cdot \sin(n\pi)\right] - \left[0 - \frac{4}{n^2\pi^2} \cdot \sin\left(\frac{n\pi}{2}\right)\right]\right\}$$

$$= \frac{2\left[n\pi \cdot \left(\cos(n\pi) + 2\sin\left(\frac{n\pi}{2}\right) - 2\sin(n\pi)\right)\right]}{n^2\pi^2}$$

$$= \frac{2\left[n\pi(-1)^n + 2\sin\left(\frac{n\pi}{2}\right)\right]}{n^2\pi^2}$$

$$u_t(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{2}\right)\left[-b_{1,n} \cdot c_n \sin(c_n t) + b_{2,n} \cdot c_n \cdot \cos(c_n t)\right]$$

Recall: $u_t(x,0) = g(x) = 0$ Thus...

$$\left\{\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{2}\right)\left[b_{2,n} \cdot c_n\right]\right\} = 0 \quad \therefore b_{2,n} = 0 \text{ so...}$$

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{2}\right)\left[b_{1,n} \cdot \cos(c_n t)\right]$$ where $b_{1,n}$ was defined above. Q.E.D.